

PROJECT PROPOSAL

OF

NUMERICAL AND EXPERIMENTAL ANALYSIS OF AN OPTIMIZED, VENTURI EFFECT WIND CONCENTRATOR FOR LOW SPEED REGIMES

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1. Project Overview

Project Title: Numerical and Experimental Analysis of an Optimized, Venturi Effect Wind Concentrator for Low Speed Regimes

Introduction: This project focuses on the development, simulation, and validation of a portable, highly optimized Diffuser Augmented Wind Turbine (DAWT). The design introduces specific fluid dynamic modifications to accelerate ambient air, effectively lowering the minimum wind speed required for operation.

Purpose and Need: Wind energy extraction in the Central Province of Sri Lanka is significantly hindered by the region's topographical layout. The Peradeniya region is characterized by complex, multi directional wind patterns and low average ambient velocities, typically ranging from 2m/s to 3m/s. Available wind power is proportional to the cube of the wind velocity, mathematically expressed as $P = \frac{1}{2} \rho A v^3$. Because of this exponential relationship, even marginal increases in airflow speed can yield massive gains in energy output. The proposed system is necessary to artificially accelerate these low-speed winds to maximize the Power Coefficient and provide a viable renewable energy solution.

2. Analysis of Existing Systems

Specifications and Limitations of Current Solutions:

- **Standard Horizontal Axis Wind Turbines (HAWTs):** Traditional HAWTs are highly inefficient in low speed conditions. They frequently fail to overcome the necessary cut in speed required to initiate power generation in 2m/s to 3m/s regimes.
- **Uni Directional Venturi Concentrators:** Standard converging diverging nozzles can successfully demonstrate velocity augmentation based on the continuity equation. However, they possess a severe limitation in mountainous topographies. They require heavy, complex mechanical yaw systems to actively track shifting wind directions. This negates the goal of maintaining a lightweight and system.
- **Standard Smooth Walled Diffusers:** Conventional diffusers are prone to boundary layer separation, which reduces the efficiency of the pressure recovery at the exit.

3. Project Objectives

To optimize the concentrator for low speed regimes, this project synthesizes four specific aerodynamic improvements verified by recent literature:

- **Omnidirectional Multi Deck Intake:** To address the shifting wind directions common in mountainous terrain, the system will utilize a static, multi deck capture area. This allows for 360 degree wind capture without requiring heavy, mechanical yaw systems, ensuring the prototype remains lightweight and portable.
- **Coanda Optimized Inlet Profile:** To maximize the mass flow rate entering the system, the inlet lip will be mathematically curved to exploit the Coanda Effect. This ensures flow attachment, maintaining a laminar air stream and drastically reducing spillage at the entry point.
- **Helical Grooved Diffuser:** Moving beyond standard smooth-walled diffusers, the internal shroud will feature helical grooves. Drawing on Yuji Ohya's "Wind Lens" research, these ribs act as vortex generators. They prevent boundary layer separation and strengthen the circumferential swirl, creating a severe low-pressure zone at the exit that forcefully draws air through the turbine throat.
- **Serrated Rotor Trailing Edges:** To maintain steady RPMs at low Reynolds numbers, the rotor blades will feature trailing edge serrations. Literature indicates that breaking up the turbulent wake at the blade tips increases the overall induction factor, reducing aerodynamic noise and allowing the turbine to operate efficiently even when wind speeds drop below 2 m/s.

4. Design Evolution and Conceptual Framework

To arrive at the final proposed architecture, the team progressed through three distinct conceptual design stages, analyzing the aerodynamic limitations at each step to inform the next iteration.

- **Stage 1: The Fundamental 3-Part Block Architecture**

The initial concept established the core functional requirements of the system. The architecture was broken down into three sequential stages:

1. **Wind Collector:** To capture ambient mass flow.
2. **Venturi Effect Turbine:** To compress the flow, increase dynamic pressure (velocity), and extract kinetic energy.
3. **Diffuser:** To recover pressure and maximize aerodynamic efficiency at the exit.

This baseline confirmed that a bare turbine is insufficient and that front end and back end augmentations are required.

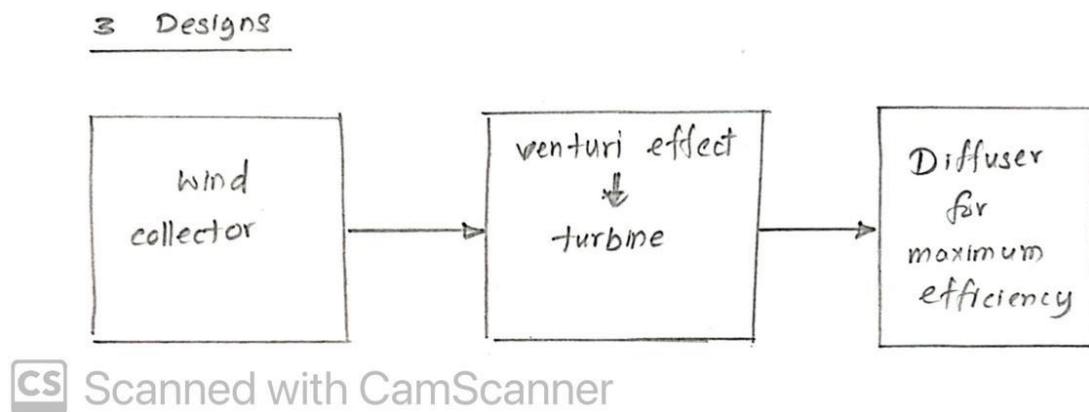


Figure 1: Block diagram of the fundamental 3-part conceptual architecture

- **Stage 2: Uni-Directional Venturi Concentrator**

The second iteration visualized the fluid mechanics in a horizontal axis configuration. This design featured a standard converging diverging nozzle with specific inlet diameters and throat diameters. While this model successfully demonstrated theoretical velocity augmentation based on the continuity equation, it presented a severe limitation for the Peradeniya topography: it required a heavy mechanical yaw system to actively track shifting wind directions.

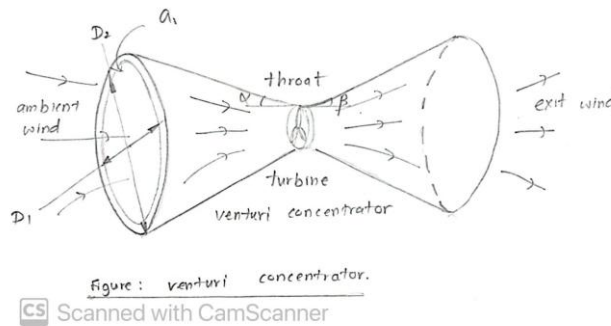


Figure 2: 2D cross-section of the Venturi Concentrator demonstrating expansion angles and throat dimensions

- **Stage 3: Omnidirectional Venturi Turbine (Final Proposed Concept)**

To eliminate the mechanical complexities and weight of a yaw system, the final design incorporates a static, multi directional (360-degree) wind collector mounted on a lightweight truss system. The captured air is channeled downwards through a narrowing vertical duct (the Venturi throat), where a 3 blade Horizontal Axis Wind Turbine (HAWT) is housed. The flow then exits through a specialized diffuser equipped with helical internal features. This omnidirectional approach maximizes the capture area and ensures continuous operation regardless of environmental wind shifts, perfectly aligning with the portability requirements of the project.

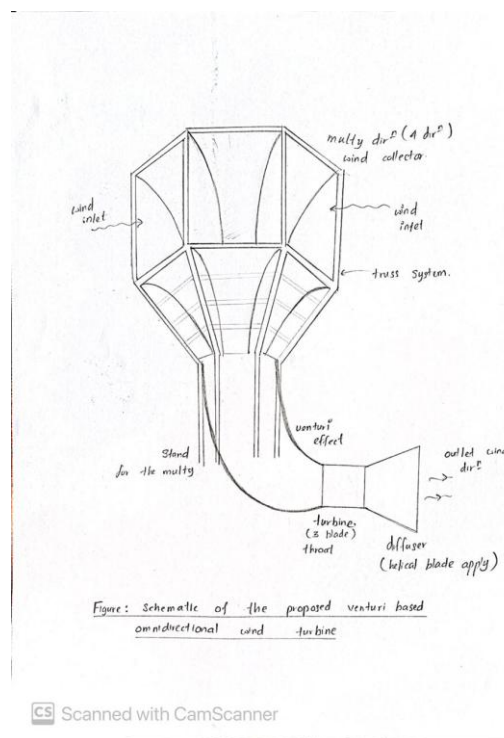


Figure 3: Schematic of the proposed Venturi-based omnidirectional wind turbine

5. Technical Methodology

The project will be executed in four distinct phases, bridging theoretical fluid mechanics with tangible lab data:

1. **Literature Baseline & Mathematical Modeling:** Utilizing the Continuity Equation and standard Betz limit theories to define the initial Area Ratio, Inlet to Throat. MATLAB will be utilized to script the theoretical power curves and velocity profiles for 2 m/s to 6 m/s wind inputs.
2. **Numerical Simulation (CFD):** High fidelity 3D CAD models will be evaluated using ANSYS Fluent. Simulations will specifically employ the turbulence model to visualize pressure gradients, wake recovery, and the vortex shedding induced by the helical grooves.
3. **Experimental Validation:** Simulate ambient winds using a standardized fan setup. Data collection will focus on the differential RPM and voltage/current output of a bare turbine versus the fully shrouded system to calculate the empirical efficiency gain.

6. Software Tools:

- **MATLAB:** For scripting theoretical power curves and velocity profiles for wind inputs
- **SolidWorks:** For translating theoretical geometries into high fidelity 3D CAD models and managing structural fabrication elements.
- **ANSYS Fluent:** For conducting Computational Fluid Dynamics (CFD) simulations

7. Project Planning & Timeline

The project timeline and workflow are structured around four defined milestones, with responsibilities distributed among the project team:

- **Milestone 1: Design Concept**
 - *Week 1 - 3*
- **Milestone 2: Numerical Analysis**
 - *Week 4 - 6*
- **Milestone 3: Modeling and Simulation**
 - *Week 7 - 10*
- **Milestone 4: Experimental Validation & Documentation**
 - *Week 11 - 14*

8. Impact Analysis

Economic and Social Considerations:

The proposed turbine offers a highly viable, portable renewable energy solution specifically tailored for low wind regions in Sri Lanka. By avoiding heavy mechanical yaw systems and utilizing lightweight polymers, the system remains easily transportable and physically accessible for rural or mountainous deployments.

Environmental Considerations:

In addition to providing clean, renewable power, the design incorporates serrated rotor trailing edges. This specific modification is intended to break up the turbulent wake at the blade tips, which increases the overall induction factor while actively reducing aerodynamic noise. Furthermore, the mathematically curved Coanda optimized inlet ensures flow attachment and maintains a laminar air stream, reducing disruptive air spillage at the entry point.

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9. References

The aerodynamic modifications proposed in this project are substantiated by the following peer-reviewed literature and experimental studies:

1. **Ohya, Y., & Karasudani, T. (2010).** *A Shrouded Wind Turbine Generating High Output Power with Wind-lens Technology*. *Energies*, 3(4), 634-649.
2. **Allaei, D., & Andreopoulos, Y. (2014).** *INVELOX: Description of a new concept in wind power and its performance evaluation*. *Energy*, 69, 336-344.
3. **Oerlemans, S., Fisher, M., & Föller, S. (2009).** *Reduction of wind turbine noise using optimized trailing-edge serrations*. *AIAA Journal*, 47(6), 1470-1481.
4. **Abe, K., Nishida, M., Sakurai, A., et al. (2005).** *Experimental and numerical investigations of flow fields behind a small wind turbine with a flanged diffuser*. *Journal of Wind Engineering and Industrial Aerodynamics*, 93(12), 951-970.